Operational Development of Marine Highways to Serve the U.S. Pacific Coast

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This paper examines the market volumes, service times, vessel characteristics, and economics for marine highways that would serve the U.S. Pacific Coast. This work was performed under contract to the Center for Commercial Deployment of Transportation Technologies. Market volumes were assessed by routes of interest and by filtering for cargo that would be eligible for marine highway service on the basis of the drayage distance and other factors. Sufficient daily truckload volumes exist in the Northern California to Southern California route to justify marine highways with multiple daily sailings of vessels with capacities of 450 to 700 trailers, should service times and economics prove to be competitive enough to divert cargo. Market volumes in the California to Pacific Northwest route are sufficient for daily sailings of smaller vessels with capacities of approximately 150 to 200 trailers. A door-to-door supply chain perspective was maintained, and discrete-event simulation was used to assess the service times and the vessel speeds required. The resulting voyage analysis served as input to both a parametric analysis of vessel characteristics and an economic analysis of marine highways by considering all maritime and land-side cost elements associated with the door-to-door movement of trailers. It was concluded that current truck rates are not high enough for marine highways to compete on the basis of cost in short next-day-turnaround markets, such as Northern California to Southern California. Marine highways are viable for longer routes such as those from California to the Pacific Northwest, where truck rates are higher and both distance and trucking hours-of-service regulations permit vessels to be time competitive at lower speeds.

This paper summarizes the approach to and findings of work performed by the authors under contract to the Center for Commercial Deployment of Transportation Technologies (CCDoTT) examining the market volumes, service times, vessels, and economics for marine highways that would serve the U.S. Pacific Coast. A door-to-door supply chain perspective that considered both land-side and maritime components was central to the study. Additional details regarding the study may be reviewed by obtaining the final report, *Operational Development of Short Sea Shipping to Serve the Pacific Coast,* from CCDoTT (*1*).

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MARKET ANALYSIS AND PERCEPTIONS

The markets of interest for this study included Northern California to Southern California and California to the Pacific Northwest. An assessment of the potential market volumes for truckloads with different levels of market penetration was conducted by using data from Transresearch Reebie. In addition, the team sought to capture trucking company and third-party logistics provider viewpoints of potential marine highways, that is, what requirements are necessary for the service to be successful, what components affect the modal decision, and what obstacles may exist for such a service. Shippers ultimately gauge transit time and price trade-offs. The survey responses represented trucking companies' and third-party logistics providers' broad knowledge of shipper price and transit time needs.

Market Volumes

The cargo flows and trade lane analysis identified 5,663 truck and rail county traffic lanes, grouped according to a broad definition of the potential market for West Coast short sea service: 14 business economic areas (BEAs) that had at least a minimal potential to be suitable for cargo diversion into the coastwise service. Northbound and southbound Pacific Coast shipments in general, with sufficient lengths of haul or origin–destination pairs that do not fall within a single port area, were generally identified as being eligible. Truck cargo was analyzed at the county level in the United States. All counties in the states of California, Oregon, and Washington were included; and the truck traffic data provided were split into three types: common carrier truckload, less than truckload, and private truckload. U.S. rail cargo, both intermodal and carload (all other types of railcars other than trailers or containers moving by intermodal car), was analyzed at the BEA level. This is the most detailed level at which cargo loads can be provided without special permission from the Surface Transportation Board. Each origin–destination pair included traffic mode (truck or rail), length of haul, and commodity type. Table 1 presents the data as daily truckloads by origin and destination. The market volumes in Table 1 are escalated from a 2004 basis to 2012. During the 15-year period from 1989 to 2004, it is estimated that truckload tonnage in California grew by about 1.6% per year on the basis of an analysis of historical California truck mileage data obtained from the California Department of Transportation.

The projected truckload volumes were examined in the context of daily sailings of large (700-trailer-capacity), medium (450-trailer-capacity), and small (150-trailer-capacity) vessels for each BEA pair. Vessels in the 150- to 200-trailer range have capacities comparable to those of overseas roll-on–roll-off (Ro-Ro) passenger ferry designs and might offer opportunities to leverage proven international designs or experience. Vessels with 450-trailer capacities

TABLE 1 Estimated Filtered 2012 Daily Truckload Flows by Origin–Destination

	Origin BEA								
Destination BEA	Los Angeles, California	San Francisco, California	San Diego, California	Seattle, Washington	Sacramento, California	Portland, Oregon	Richland, Washington	Other	Total
Boise City, Idaho	4	0	0	0	0	0	0	0	4
Eugene, Oregon	128	33	7	736	16	222	5	4	1152
Fresno, California	0	0	46	91	0	70	339	67	613
Los Angeles, California	530	25,886	5,000	1,607	4,902	1,506	401	1,589	41,420
Pendleton, Oregon	324	7	1	0	2	0	0	2	334
Portland, Oregon	370	518	30	1,435	84	0	0	494	2,931
Redding, California	789	0	78	145	0	13	0	0	1,025
Reno, Nevada	63	0	25	1	0	2	0	0	92
Richland, Washington	64	64	2	0	9	0	0	25	164
Sacramento, California	4,801	0	516	107	0	92	64	63	5,643
San Diego, California	3,691	2,656	0	107	456	95	16	168	7,189
San Francisco, California	26,097	0	2,889	869	0	478	962	397	31,692
Seattle, Washington	863	354	37	911	100	842	0	347	3,456
Spokane, Washington	121	35	3	0	10	0	0	9	178
Grand Total	37,844	29,553	8,634	6,009	5,579	3,321	1,788	3,165	95,893

are comparable in size and configuration to Totem Ocean Trailer Express's newest vessels, although the California market will require higher speeds. Vessels with capacities of 700 trailers represent the largest short sea shipping vessels that are considered to be serviceable in the terminals of interest. Table 2 shows the number of daily sailings, one way, for vessels of these capacities in each of the markets under the assumption of various levels of market penetration and 312 transportation days per year. This represents 52 weeks of 6 days per week, consistent with U.S. Environmental Protection Agency assumptions of 300 truck days per year for the purpose of calculating idling emissions (*2*). To account for balance, Tables 2 through 4 reflect the lower of the two volumes north or south. The BEA pairs represented in the figure include Los Angeles, California–San Francisco, California; Los Angeles–Seattle, Washington; San Francisco–Seattle; Oregon–Seattle; Sacramento, California–Los Angeles; San Diego, California–Los Angeles; and Los Angeles–San Francisco with Sacramento. The truckloads in the Sacramento–Los Angeles market would really be additive with truckloads in the Los Angeles–San Francisco market, with drayage occurring between San Francisco and Sacramento.

Trucking and Logistics Company Perceptions

Representatives of 20 West Coast trucking companies and third-party logistics providers selected by TranSystems and Westar Transport were interviewed in May and June 2007. The companies interviewed did see a role for marine highways in certain situations, provided that service requirements were met. The following are highlights of the commodity characteristics and service requirements put forth by the survey respondents:

- Trucking companies believed that a marine highway service was more appropriate for distances greater than 700 or 800 mi for non-time-sensitive cargoes. Shorter transit distances, such as the distance from Los Angeles to the San Francisco Bay Area, are considered to be overnight markets (defined as 8 to 12 h of truck transit time) and not appropriate for a coastwise shipping option. Low-margin, non-time-sensitive cargoes might be eligible, however, if price discounts were offered.
- Trucking companies indicated that transit times of 1 to 2 days between Los Angeles and San Francisco and 2 to 3 days between Los Angeles and Seattle would be required in most cases.
- Discounts of 20% to 30% off trucking costs, in general, are considered to be sufficient to compensate for a transit time increase of 1 day for longer short sea transits, assuming that the coastwise service is reliable.
- Respondents most often suggested the use of daily sailings to reduce the time spent waiting for the ship to sail. Drayage to the port, plus waiting for vessel sailing times, are viewed as incremental time to a coastwise shipping option; and the respondents believed that these additional delays could result in transit times that were two to three times longer than truck transit times.
- Commodities that might be eligible to be shifted offshore are either non-time-sensitive, low-value cargoes that are being used for non-just-in-time warehouse replenishment or materials for manufacturing for which longer transit times have been built into the supply chain.
- A marine highway service is more viable for large trucking companies that have a broad geographic scope and that have tractors in both origin and discharge ports. Owner–operator trucking companies, which typically operate from a single home base, would not receive an advantage from marine highways because they typically would not have the ability to arrange for the pickup of cargo at the discharge port. The short sea shipping program should include an owner–operator network to coordinate owner–operator handoffs at load and discharge ports. A trailer equipment pool that would provide owners–operators with replacement trailers during the duration of the round-trip voyage was also recommended.
- The Pacific Northwest freight imbalance may create the need to reposition trailers that are empty on the lower-demand southbound leg.

TABLE 2 Estimated 2012 Daily Sailings by BEA Pair, One Way

Market Share	LA–SF	LA–SEA	SF–SEA	OR–SEA	SAC–LA	SD–LA	LA–SF/SAC
700 Trailer Vessel							
2%	1	0	0	0	0	0	1
4%	2	0	0	0	0	0	2
6%	3	0	0	0	0	0	4
8%	4	0	0	0	0	0	5
10%	5	0	0	0	1	0	6
12%	6	0	0	0	1	0	8
14%	8	0	0	0	1	1	9
16%	9	0	0	0	1	1	10
18%	10	0	0	0	1	1	12
20%	11	0	0	0	2	1	13
22%	12	0	0	0	2	1	15
24%	13	0	0	0	2	1	16
26%	14	0	0	0	2	2	17
28%	16	0	0	0	2	2	19
30%	17	0	0	0	3	2	20
32%	18	0	0	0	3	2	21
34%	19	0	0	0	3	2	23
36%	20	0	0	0	3	2	24
38%	21	0	0	0	4	3	25
40%	23	0	0	0	4	3	27
42%	24	0	0	0	4	3	28
44%	25	0	0	0	4	3	30
46%	26	0	0	0	4	3	31
48%	27	0	0	0	5	3	32
50%	28	0	0	0	5	4	34
450 Trailer Vessel							
2%	0	0	0	0	0	0	0
4%	1	0	0	0	0	0	1
6%	2	0	0	0	0	0	2
8%	2	0	0	0	0	0	3
10%	3	0	0	0	0	0	4
12%	4	0	0	0	0	0	5
14%	5	0	0	0	0	0	6
16%	5	0	0	0	1	0	7
18%	6	0	0	0	1	0	7
20%	7	0	0	0	1	1	8
22%	8	0	0	0	1	1	9
24%	8	0	0	0	1	1	10
26%	9	0	0	0	1	1	11
28%	10	0	0	0	1	1	12
30%	11	0	0	0	2	1	13
32%	11	0	0	0	2	1	14
34%	12	0	0	0	2	1	14
36%	13	0	0	0	2	1	15
38%	14	0	0	0	2	2	16
40%	14	0	0	0	2	2	17
42%	15	0	0	0	2	2	18
44%	16	0	0	0	3	2	19
46%	17	0	0	0	3	2	20
48%	17	0	0	0	3	2	21
50%	18	0	0	0	3	2	21

(continued)

TABLE 2 *(continued)* **Estimated 2012 Daily Sailings by BEA Pair, One Way**

Market Share	LA–SF	LA–SEA	SF–SEA	OR–SEA	SAC–LA	SD–LA	LA–SF/SAC
150 Trailer Vessel							
2%	3	0	0	0	0	0	4
4%	6	0	0	0	1	0	8
6%	10	0	0	0	1	1	12
8%	13	0	0	0	2	1	16
10%	17	0	0	0	3	2	20
12%	20	0	0	0	3	2	24
14%	24	0	0	0	4	3	28
16%	27	0	0	0	5	3	32
18%	31	1	0	1	5	4	36
20%	34	1	0	1	6	4	40
22%	37	1	0	1	7	5	45
24%	41	1	0	1	7	5	49
26%	44	1	0	1	8	6	53
28%	48	1	0	1	8	6	57
30%	51	1	0	1	9	7	61
32%	55	1	0	1	10	7	65
34%	58	1	0	1	10	8	69
36%	62	2	0	2	11	8	73
38%	65	2	0	2	12	9	77
40%	69	2	0	2	12	9	81
42%	72	2	0	2	13	10	85
44%	75	2	1	2	14	10	90
46%	79	2	1	2	14	11	94
48%	82	2	1	2	15	11	98
50%	86	2	1	2	16	12	102

NOTE: LA = Los Angeles; SF = San Francisco; SEA = Seattle; OR = Oregon; SAC = Sacramento; SD = San Diego.

- Additional anticipated benefits from marine highways, such as reduced traffic congestion, reduced truck pollution, improved working conditions for truck drivers, and alleviation of the truck driver shortage were acknowledged; but marine highways received only moderate support from the trucking companies that responded to this survey. The potential cost savings was the key advantage when they considered using a coastwise service.
- The responses suggested that diverting over-the-road truck volume into port areas for a marine highway service may compound existing traffic congestion issues in and around the ports.

DOOR-TO-DOOR SIMULATION MODEL

An in-port discrete-event simulation model was developed to support an analysis of the infrastructure requirements and vessel speeds required for marine highway operations on specific routes. The simulation determined the time required to complete a cycle of unloading and loading a vessel, as well as pickups and deliveries, given assumptions regarding the available resources, terminal characteristics, vessel characteristics, and market parameters, such as drayage distances. With this information for both the southern and the northern terminals, a voyage analysis could be performed to determine the door-to-door delivery standard that may be met for a given vessel cruising speed. Figure 1 illustrates the components of such a voyage analysis, in which vessel speeds would vary depending on the portion of the route. Figure 2 also illustrates the major elements of the in-port discrete-event model that addresses the drayage to and from and terminal portions of the voyage analysis. The in-port model was developed in the EXTEND simulation environment, with the remainder of the voyage analysis addressed by using the output of the discrete-event model within a model built in Excel software.

The key variables that may be the subject of experiments in this simulation model include the following (variables treated as stochastic probability distributions are so indicated):

- Number of yard tractors,
- Number of drayage tractors,
- Trailer capacity,
- Number of inbound lanes,
- Number of outbound lanes,
- Inbound gate delay (stochastic),
- Outbound gate delay (stochastic),
- Drayage distance between terminal and receiver (stochastic),
- Drayage distance between terminal and shipper (stochastic),
- Truck speed (stochastic),
- Delays in processing trailers at shippers (stochastic),
- Delays in processing trailers at receivers (stochastic),
- Distance between receiver and shipper for return loaded trip (stochastic),

TABLE 3 Marine Highways Vessel Characteristics

Vessel Parameter	Unit	450 Trailer, 24 knots	550 Trailer, 27 knots	700+ Trailer, 27 knots	500–600 Trailer SL-7, Variant, 32 knots	150 Trailer, 22 knots	150 Trailer, 24 knots	150 Trailer, 26 knots
Length overall (LOA)	Feet	832.5	837.7	959.6	980.1	484.16	484.16	509.16
Gross registered tons (GRT)	Long tons	38,050	39,068	40,600	39,068	15,875	15,875	15,875
Breadth overall (BOA)	Feet	118.0	118.0	118.1	105.5	69.85	74.22	71.43
Length (waterline) (LBP)	Feet	794.4	794.4	904.4	900.0	475.0	475.0	500.0
Beam (waterline) (BWL)	Feet	118.0	118.0	118.1	105.5	69.85	74.22	71.43
Draft	Feet	19.84	23.66	26.70	23.41	17.25	17.25	19.92
Depth	Feet	81.40	93.70	102.70	79.91	42.43	42.43	45.1
Displacement, full ship	Long tons	31,140	37,020	47,845	33,722	8,717	9,239	10,851
Displacement, light ship	Long tons	20,743	24,313	26,173	21,668	5,666	4,645.63	5,830
Design speed	Knots	24	27	27	32	22	24	26
Power, installed	SHP	69,954	119,829	121,536	117,768	19,693	28,800	41,054
Length over beam (L/B)	6.732	6.732	7.660	8.531	6.800	6.400	7.000	
Block coefficient (CP)	0.691	0.671	0.660	0.577	0.589	0.589	0.589	
Payload	Long tons	10,397	12,707	16,588	12,054	3,894.44	4,128.5	4,427.76
Payload	Trailers	450	550	718	550	150	150	150
Hull decks		5	6	7	5	2	2	2

NOTE: SHP = shaft horsepower; SL-7 = T-AKR 287 Algol SL-7 class of fast sealift vessels.

TABLE 4 Costs per Trailer, Southern California to Northern California Lane, 2008

Cost Component	Cost per Trailer Slot–2008 Low ($)	High ($)
Capital (annual loan payment)	183	224
Bunker (fuel)	294	529
Crew	39	39
Food and stores	1	1
Ship insurance	5	5
Repairs and maintenance	4	4
Navigation and port charges	55	55
Trailer stevedoring (2 handles)	300	400
Dray to and from the port	440	500
Other miscellaneous expenses	20	26
Total	1,341	1,783
Estimated average trucking cost to shipper (per trailer)	950	950

- Number of inbound trailers prestaged in the terminal before the start of unloading or loading,
- Vessel throughput in number of trailers per hour for loading (stochastic), and
- Vessel throughput in number of trailers per hour for unloading (stochastic).

The output of the model includes

- Number of trailers unloaded over time,
- Number of trailers loaded over time,
- Number of trailers delivered to receivers over time,
- Staging required for inbound and outbound trailers,
- Total time required to process a vessel,
- Time from unloading to delivery, and
- Time from pickup to loading.

A number of the input variable probability distributions, such as threshold and objective vessel throughput rates, were derived from observation of existing Ro-Ro operations. TransSystems developed drayage distance distributions for the BEA pairs of interest. Figure 2 illustrates the distribution for Los Angeles to San Francisco. In the experiments conducted, schedules were predicted by using both the full distribution and an abbreviated distribution capping the maximum drayage distance to determine the impact of being more selective about the choice of freight.

The U.S. Department of Transportation maintains statistics regarding average truck speeds along 10-mi segments of routes of interest, such as I-5 (*3*). Along the entire length of I-5, the average speed is 48.43 mph. In the vicinity of major centers, such as Los Angeles, average speeds are as low as 40 to 45 mph. The mode was assumed to be 45 mph, with a minimum of 35 mph and a maximum of 50 mph being assumed.

The processing of a load at either the shipper or the receiver would be expected to be completed within 15 min. However, occasional delays would be expected to occur. Delays have been modeled with the potential for up to 1 h in one case and 30 min in another to explore the impact of controlling this variable on overall service times.

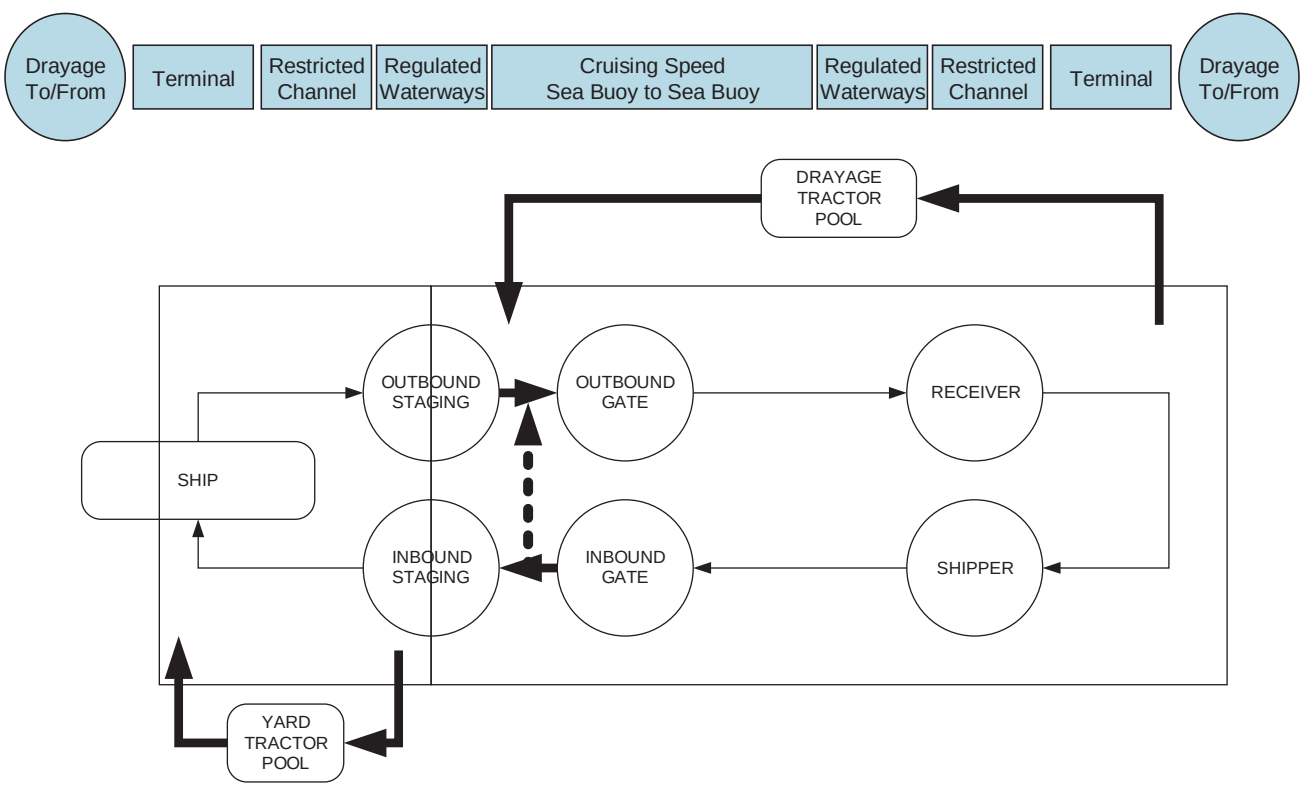

FIGURE 1 Door-to-door model and simplified in-port simulation model structure.

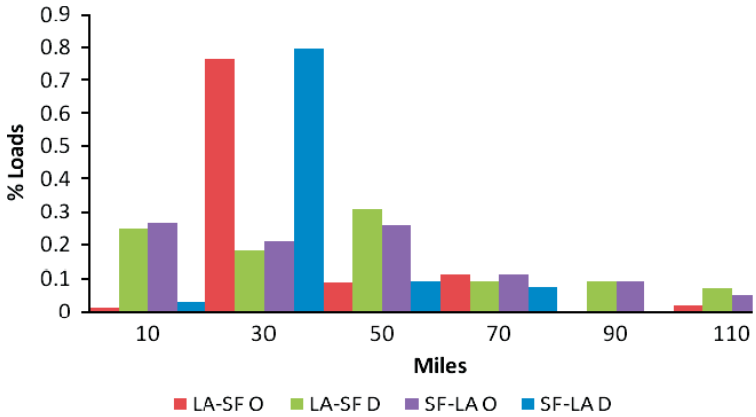

FIGURE 2 Drayage distances for Los Angeles and San Francisco (O = origin; D = destination).

For the 450-trailer vessel on the Northern California to Southern California route, it was determined that the following terminal parameters were consistent with the next-day service required:

- 15 inbound and outbound gates,
- 350 drayage tractors, and
- 50 yard tractors.

In a scenario designed to capture a broad range of freight with drayage distances of up to 70 mi at the Los Angeles end and up to 110 mi at the San Francisco end, it was determined that staging would begin in Los Angeles approximately 2 h before vessel unloading and in San Francisco approximately 2.3 h before vessel unloading. With a cruising speed of 27 knots, delivery times ranged from 23.4 to 24.8 h. In this scenario, at 27 knots an operator would need to obtain a percentage of the freight that was less time sensitive and that could be delivered later than 24 h after pickup. At 28 knots, the most probable delivery time is less than 24 h. In this scenario, staging for up to approximately 300 trailers is required at the Los Angeles end and for up to approximately 350 trailers at the San Francisco end.

If the freight was selected to reduce the maximum drayage distance such that only freight 10 mi over the most probable distances was selected (a maximum of 40 mi for Los Angeles and a maximum of 60 mi for San Francisco), then the vessel speed may be reduced to 26 knots. In this scenario, staging in both Los Angeles and San Francisco begin 1.6 h before vessel unloading. In this scenario, staging for up to 280 trailers is required at the Los Angeles end and for up to 260 trailers at the San Francisco end.

Reducing vessel throughput from the objective to the threshold (25% reduction) broadens the range of delivery times, with the maximum and most probable times being modestly increased. Vessels would need to be designed to maximize throughput through the use of multiple ramps with access to multiple decks and an efficient terminal interface. If an operator is able to coordinate well between deliveries and pickups to minimize the distance traveled between the two by drayage tractors, then the delivery times can be improved. Maximum times can be reduced by limiting this deadhead distance to a mode of 15 mi and a maximum of 20 mi rather than distances comparable to terminal drayage distances.

Figure 3 provides an overall comparison of the four 450-trailer-capacity scenarios explored for the Los Angeles to San Francisco route on the basis of the maximum mode delivery times. The maximum mode is the maximum of the most probable delivery times for the first pickup to the first delivery or the last pickup to the last delivery for either the Los Angeles or the San Francisco end of the route. Presented are the baseline scenario, the scenario modified to reduce the drayage distance, the reduced-drayage-distance scenario further modified to use the threshold rather than objective vessel throughput rates, and the reduced deadhead distance scenario.

VESSEL CHARACTERISTICS

The characteristics of Ro-Ro vessels with capacities of 700, 450, and 150 trailers that met port access requirements for terminals in the California and Pacific Northwest markets of interest were explored by using the proprietary COMPASS design synthesis software of the Band Lavis Division of CDI Marine Company. The characteristics of the vessels identified are summarized in Table 3. First, the parametric study for the large 700-trailer vessel conducted in Phase I was rerun by using updated information on the trailer weight and with mix of commodities, with particular note of the special needs of trailers containing hazardous materials and tankers being taken. By using the characteristics presented in Table 3, an initial hull form was constructed in the FastShip program. This hull form was then exported to the Rhino3D program, in which the hull was given a more boxlike shape above the waterline to accommodate as many trailers as possible without affecting hydrodynamic performance or compromising stability. Figure 4 illustrates the resulting configuration. A stability analysis was run to establish watertight subdivision boundaries to try to improve the load-out arrangements and traffic flow. To augment this study, a visit to the Tote facility was made to see how this type of operation works in the real world. Parametric studies were made for the medium and small vessels to better understand their characteristics. Lastly, by using the results of the parametric studies for the large and small vessels, a search of Lloyds Ship Information Database Registry was performed to identify potential existing ships with proven performance and operating costs that could be used for the intended service.

It should be emphasized that identification of the ports involved is the key constraint in determining the characteristics of the vessel. The port defines

- The physical constraints, based on location, placed on the vessel design, that is, the navigational draft, beam, and length, which determine the maximum potential payload for a maximum-sized ship;

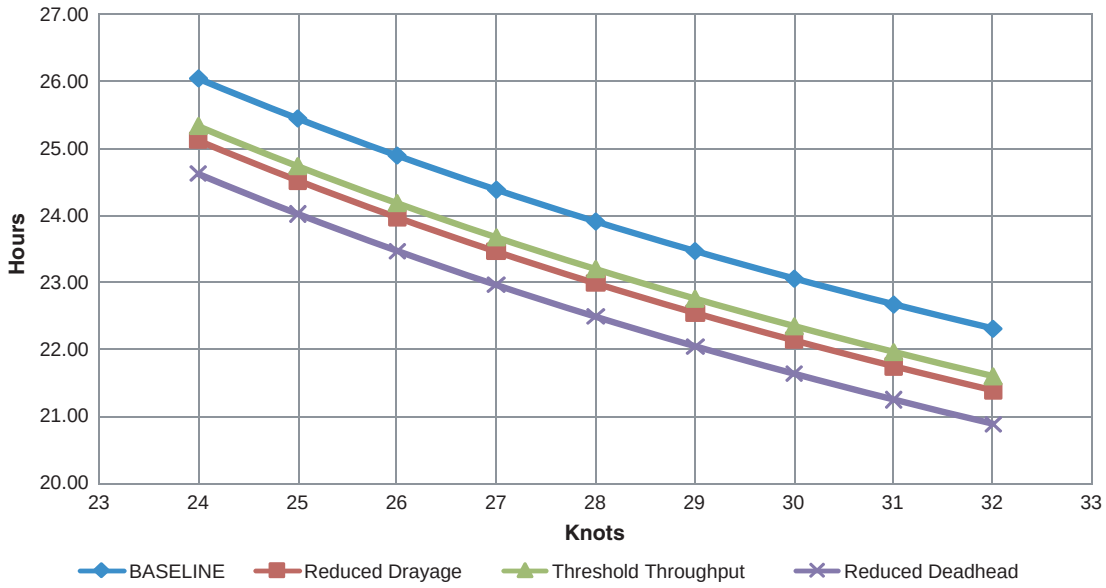

FIGURE 3 Vessel speed versus delivery times for a 450-trailer vessel on the Los Angeles–San Francisco route.

- The numbers and sizes of thrusters and the number of tugs required so the ship can be maneuvered through the channels and at the port; this also determines the degree to which pilots and independent tug companies must be employed;
- The size of the staging area, which, in turn, dictates the shoreside ramp configuration for the ship and the staging area for the trailers and their ability to move in and out of the port; and
- The maximum speed of the vessel, which, in turn, determines the size, acquisition cost, and operating cost of the propulsion plant; this is a result of how far the port is with respect to the sea buoy and distances that must be traveled in restricted or regulated waterways.

ECONOMICS OF MARINE HIGHWAYS

The cost components related to marine highways are outlined in Table 4. Additionally, some other cost components that currently apply to all cargoes that use U.S. ports, such as the harbor maintenance tax and the merchandise processing fee, must be considered to estimate the monetary impact on marine highway costs to the shipper.

The notional vessel that is the focus of this economic review is a design hybrid between the 450-trailer vessel traveling at 24 knots and the 550-trailer vessel traveling at 27 knots described in Table 3. The deployment reviewed is the Southern California to Northern California lane, the lane with the largest potential volume analyzed in this study. If a vessel cruise speed of 27 knots and a payload of 450 trailers are assumed, this deployment would require two vessels, if daily service was offered. This allows a schedule contingency of an estimated 7%. This two-vessel string would have a cargo receipt cutoff of late afternoon, would sail the 16 h during the night, and would arrive in the early morning hours to be unloaded for the delivery of cargo throughout the day, providing close to 24-h service for a certain portion of the cargo. For the single leg of the journey, the estimated time in port is 7 h and the estimated sailing time is 16 h (380 nautical miles at an average overall speed of 24 knots resulting from restricted and regulated waterways), providing almost 1.5 h of contingency time in the schedule. This timing is consistent with the discrete-event modeling performed in previous tasks.

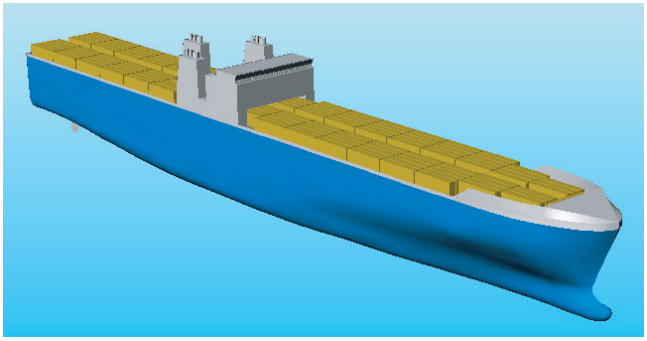

FIGURE 4 Large 700-trailer Ro-Ro vessel.

Table 4 summarizes the analysis associated with the financial competitiveness of a marine highway service versus a truck mode on the U.S. West Coast moving domestic cargo between Southern California and Northern California. The multitude of regulatory requirements associated with this Jones Act trade, as well as the elements of the proposed business model attributes, results in costs that are currently not competitive against trucking costs. These costs, on a per-trailer-slot basis, are displayed, along with the current average truck cost that shippers pay. Low- and high-cost outlooks are provided. These are based on varying the different underlying assumptions for the major cost-contributing components.

The largest contributors to the total cost are the fuel, drayage, and stevedoring components, which account for approximately 80% of the total per trailer cost. The capital cost of the vessel is a smaller contributor to the overall operating cost of moving cargoes via this mode. Current indications of truck rates on the Southern California to Northern California route are that southbound rates range from $800 to $900 and northbound rates range from $950 to $1,100. These rates are inclusive of the current estimated fuel surcharge of some 35% (as of April 2008).

Construction costs were estimated by using cost-estimating relationships developed by CDI Marine Company and were incorporated

into its parametric design synthesis model, COMPASS, escalated to 2008 dollars. Estimated vessel procurement costs varied from $260 million to $410 million, depending on the vessel's capacity and speed. The costs estimated by COMPASS were consistent with the published prices and the costs of Ro-Ro vessels built in the United States from 1983 to 2003 escalated to 2008 dollars. The notional vessel reflected in Table 4 is estimated to cost $300 million in 2008 (averaging the costs of the 24-knot and the 27-knot vessels), and 100% of the cost of the vessel is expected to be financed over 30 years. The estimated annual cost was calculated by using two interest rates, one of which was associated with the Title XI program and the other of which was a commercial rate. Although the Title XI program is not currently available, the analysis includes this calculation in the event that a similar program may be initiated to encourage domestic shipbuilding.

The round-trip voyage from Southern California to Northern California includes various types of waterways that demand a specific vessel speed, thus having an impact on fuel burn. In this analysis, it is assumed that while the vessel is in port, the vessel will use shore power, and this charge is included under navigation and port charges in Table 3. The number of barrels of fuel consumed were based on the specific fuel consumption for the amount of time and the amount of power required at each speed, plus the fuel consumption associated with ship services (assumed to be constant across the profile for this analysis for each point design). Both propulsion power and ship services were predicted by using COMPASS. The calculation of fuel cost assumes that the California Air Resources Board Main Engine Fuel Rule is in effect. That rule requires the use of marine gas oil or low-sulfur medium diesel oil within 24 mi of the coast. The assumption of an approximate total cost of $521 per metric ton ($0.24/lb) and $1,021.5 per metric ton ($0.46/lb) for intermediate fuel oil and medium diesel oil, respectively provides a basis for an overall calculation of the total fuel cost per trip. A higher cost was also calculated to represent a scenario in which only the cleaner-burning medium diesel oil was used.

The vessels on this deployment will be operating under the regulatory provisions of the Jones Act with a U.S. crew. For economic modeling purposes, it is assumed that the vessels will be manned by 10 officers and 12 crew, on the basis of a review of the current crew sizes of other operators on Jones Act trades. The estimated wage rates, inclusive of benefits, are $150,000 per year and $255,000 per year for crew and officers, respectively.

Food and stores refers to the items necessary to outfit the vessel for the efficient maintenance and running of the ship as well as its crew. This inventory would include food for the crew as well as spare parts for simple repairs. The estimated annual cost for this expense component is $150,000 and is based on sources ranging from $90,000 to $160,000.

Insurance costs are estimated to be $550,000 per vessel per year and are based on estimates of other Jones Act vessel operators that range from $360,000 to $550,000. This estimate conservatively covers both the hull and the machinery and includes protection and indemnity.

The maintenance and repair cost is a function of the type, size, and age of the vessel; the type of propulsion equipment used; and the route on which the vessel is deployed. The routes in which these vessels will be deployed, particularly with the frequency of port calls, provides ample opportunities for regular maintenance. The sample of costs from other Jones Act operators suggests that the cost of vessel maintenance and repair ranges from $350,000 to $1,200,000 per year. Given the variables influencing the repairs and maintenance for this particular deployment, an estimate of $500,000 is being used.

Navigation and port charges include expenses for tugs, pilotage, and port usage fees, such as dockage and wharfage fees. The wharfage charge, assessed on the basis of the value of the cargo, is a pass through from the shipper to the port authority, but this is an expense that would not be incurred by a shipper using an alternative mode such as truck or rail. The economic model uses the current tariffs for Los Angeles and Richmond, California.

It was assumed in the results reflected in Table 4 that the discharge and loading of trailers onto the short sea shipping Ro-Ro vessel would be done by a third party. This simplifying assumption eliminates the theorizing related to land value and availability in the ports of preferred operation. It also eliminates the need for an extremely large capital outlay in the first few years for terminal infrastructure and cargo-handling equipment. This third party, a terminal operator, is likely already well established in the ports of interest and can offer rates that are more economical to the short sea operator, as this operation will initially use only a portion of the facility. That being said, the likely Ro-Ro handling rates on a per unit basis are estimated to range from $150 to $200 per move, including the gate charge but excluding wharfage. These estimates are based on input from stevedoring companies along the West Coast. The final amount will be dependent on a number of factors, such as vessel design and capacity and the annual volume commitment made to the terminal.

In addition to the cost of moving the cargo on the vessel, there is the cost of delivering the cargo to or from the terminal, known as "drayage." In this scenario, the short sea shipping service provides door-to-door service, making the mode of transport used transparent to the customer. The precise cost of drayage is dependent on the distance from the port. This model assumes that cargo is picked up or delivered within 50 mi of the port. It also assumes that a new load will be picked up on the way back to the port and travel only an additional 10 mi. For modeling purposes, this round-trip of 110 mi is estimated to cost $250, and assuming that a matched load can be picked up, this amount is discounted, resulting in an estimated $220 per dray. Drayage operators typically quote a round-trip rate because the matching of loads is difficult. Each trailer requires a dray on each end of its journey, so the total drayage cost per trailer is estimated to be $440. This drayage estimate is based on current rates for drayage on containerized cargo moving in and out of Northern and Southern California terminals. These rates may reflect the congestion faced in some of these terminals, and thus, depending on which terminals ultimately handle the short sea shipping service, these rates may be slightly overstated.

The expenses for administrative personnel would also need to be included. These management, general, and administrative costs would cover expenses related to management, marketing, finance–accounting, legal work, and human resources. Typically, within specific industries these costs account for a fairly consistent share of the total revenue. This percentage is a function of the size of the organization and any economies of scale that may exist within the organization. Not knowing the exact operating circumstances restricts the authors from making a precise percent estimate, although for modeling purposes, 1.5% of total expenses instead of revenue is used, as revenue is unknown at this time. Because each cost item will be shown separately, the results will allow the assessment of competitiveness both with and without this expense.

Other charges that are part of a shipper's expense when a seaborne mode is used are wharfage and the harbor maintenance fee (HMF). The HMF is an ad valorem tax and is currently assessed on imports, domestic cargo, and passengers at 0.125% of the declared value (or, in the case of passengers, their ticket value). Wharfage is the fee for

moving cargo across the wharf levied by the port authority and is based on each port's unique tariff and the weight (or cubic meters, etc.) of the cargo.

CONCLUSIONS AND RECOMMENDATIONS

The most significant cost contributors to marine highways are drayage, trailer stevedoring, and fuel. Vessel procurement cost is the fourth significant cost contributor. The Northern California to Southern California market is a challenging route for marine highways. This relatively short (400-mi) route is characterized by next-day service. As a result, the vessel speeds required are high (26 to 28 knots), resulting in high fuel costs and increased procurement costs. The procurement costs per trailer are minimized by a operating a larger vessel; however, larger vessels take longer to load and unload and therefore require higher speeds. Although a model that uses 450-trailer vessels does not currently look economically feasible for attracting cargo from its current truck mode, smaller vessels operating at greater frequencies at reduced speeds may prove to be viable if the savings in fuel costs are sufficient to offset the increased procurement, crew, insurance, and maintenance costs per trailer. Smaller vessels may prove to be more suitable for a greater number of terminal facilities. The markets from California to the Pacific Northwest are on substantially longer routes, and the daily trailer volumes are potentially less. On these routes, smaller vessels with capacities of 150 trailers would be appropriate. Rather than being next-day markets, these are 2- and 3-day markets that are more greatly affected by trucking hours-of-service regulations. The required vessel speeds in the Pacific Northwest routes would range from approximately 20 to 26 knots. On these longer routes, with reduced fuel consumption rates, marine highways are anticipated to be cost competitive with trucks. It is unclear how truck rates may increase relative to marine highways costs in the future. Should truck rates increase as a result of reduced capacity, increased demand, changes to hours-of-service rules, increased taxes and fees to address infrastructure costs, or higher fuel surcharges, marine highways may become cost competitive in California. In addition, the assumption of next-day service should be reexamined with potential shippers. If a sufficient volume of trailers that could be delivered more than 24 h from the time of pickup could be obtained, then vessel speeds could be reduced, which would reduce fuel costs.

Some potential options that should be considered include the use of

- An operation that attracts less time-sensitive cargo at reduced cost,
- An operation that employs its own drayage personnel exclusively to pick up and deliver cargo as a means of reducing drayage costs,
- An operation that loads and unloads vessels with its own labor or that has negotiated reduced stevedoring charges,
- A service that trades frequency for speed at lower capital and operating costs,
- A more fuel efficient vessel design, and
- A vessel that incorporates technology to speed loading and discharge and arrival and departure.

The following suggestions were received from trucking companies to make marine highways more viable:

- Marine highway ocean freight rates should be low enough to reduce overall door-to-door transportation expenses.
- Create an owner–operator network that connects owners–operators at load and discharge ports.
- Provide trailers as part of the service.
- Make the load and discharge process at the ports fast and easy.
- Ensure excellent communication between the vessel and the trucking companies so that shippers can be kept aware of estimated delivery times.

In general, trucking companies were receptive to an alternative method of transportation. Service would necessarily need to be reliable, although depending on the commodity type, the transit time may be flexible, as long as the cost–transit equation results were positive. Cost appeared to be the most important component in considering alternate modes of transport.

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